ES96T: Advanced Wireless Systems and Networks

Advanced Wireless Systems and Networks Assignment

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Task 1: Channel Fading in Wireless Communications

Task 1.1: Pathloss Calculation

The pathloss is calculated using the Hexagonal cell layout model for the Urban Micro (UMi) NLoS scenario (Molisch, 2011). The pathloss expression is:

$$PL = 36.7\log_{10}(d) + 22.7 + 26\log_{10}(f_c)$$

Where f_c is the carrier frequency in GHz and d is the distance in metres. Substituting the given values $f_c = 1$ GHz and $d = 100$ m from the briefing gives:

$$PL = 36.7\log_{10}(100) + 22.7 + 26\log_{10}(1)$$
$$PL = 36.7 \times 2 + 22.7 + 26 \times 0$$
$$\boldsymbol{PL = 96.1\ \mathrm{dB}}$$

Task 1.2: Coverage and Outage Probability

The received power is given by (Rappaport, 2002):

$$P_r = P_t - PL + X_\sigma$$

Where P_t is the transmit power in dBW, PL is the pathloss in dB and $X_\sigma \sim \mathcal{N}(0, \sigma^2)$ represents log-normal shadow fading. With $P_t = 1\ W = 0\ dBW$ and $PL\ =\ 96.1\ dB$ from Task 1.1, the mean received power is

$$\bar{P_r} = P_t - PL = 0 - 96.1 = -96.1\ \mathrm{dBW}$$

The shadow fading standard deviation is $\sigma = 4$ dB as specified for the hexagonal cell layout model (Molisch, 2011). Thus,

$$P_r \sim \mathcal{N}(-96.1,\ 4^2)$$

Coverage probability is defined as the probability that the received power exceeds the minimum acceptable level of -104.1 dBW:

$$P_c = P(P_r \geq -104.1)$$

Convert to standard normal:

$$Z = \frac{-104.1 - (-96.1)}{4} = \frac{-8}{4} = -2$$

$$P_c = P(Z \geq -2) = 0.9772$$

$$P_{out} = 1 - P_c = 0.0228$$

So final answers are obtained as $\boldsymbol{P_c = 0.9772}$ and $\boldsymbol{P_{out} = 0.0228}$. The coverage probability of Pc = 97.72% exceeds the typical industry target of 95% (Ofcom, 2024), indicating satisfactory coverage at 100 m from the base station. The outage probability of 2.28% reflects the residual effect of shadow fading (Rappaport, 2002), which causes occasional signal drops below the minimum threshold despite the mean received power being 8 dB above it.

Task 1.3: MATLAB Implementation

The pathloss model from Task 1.1 and the received power calculation from Task 1.2 are implemented in the MATLAB function hex_pathloss(fc, Pt, d, sigma). The function returns received power in dBW, with shadow fading drawn as a zero-mean Gaussian random variable of standard deviation σ in the dB domain. Calling hex_pathloss(1, 1, 100, 0) with $\sigma = 0$ produces −96.10 dBW, confirming the implementation is consistent with Task 1.1.

Task 1.4: Simulation of Shadow Fading

A Monte Carlo simulation was implemented using 1000 independent realisations of the shadow fading component, generated by calling hex_pathloss(1, 1, 100, 4) repeatedly. For each realisation, the received signal power was computed and compared against the minimum acceptable signal level of −104.1 dBW. Coverage is determined by finding the percentage of signals received that were above this level.

The simulated coverage probability was computed as **PC_sim = 0.9820 (98.20%)**. The simulated result is in close agreement with the theoretical coverage value of **PC_Theory = 0.9772 (97.72%)** derived in Task 1.2, with a difference of only 0.48%. The small discrepancy is expected due to the finite number of realisations (N = 1000). With more realisations the simulated value would converge to the theoretical value.

The histogram of received signal power is shown in Figure 1, with the coverage threshold of −104.1 dBW indicated by the red dashed line. The distribution is approximately Gaussian in the dB domain, centred around the mean received power of −96.1 dBW, consistent with the lognormal shadowing model.

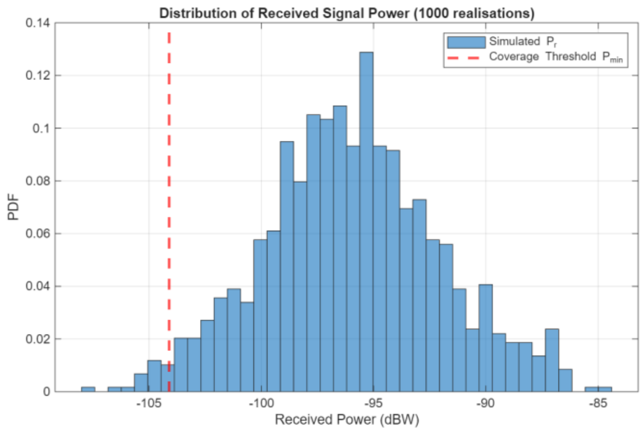

Figure 1: Histogram showing distribution of received signal power over 1000 realisations

Task 2: Radio Resource Allocation in 4G Cellular Networks

Task 2.1: Simulation Setup

A two-dimensional simulation space was constructed consisting of two adjacent 1km x 1km cells. A base station was placed at the centre of each cell at coordinates $BS_1 = (0.5, 0.5)$ km and $BS_2 = (1.5, 0.5)$ km. A single user terminal (UT) per cell was positioned according to a uniform random distribution within the cell boundaries using MATLAB's rand() function, simulating realistic user placement. The site locations are isualised in Figure 2, where triangular markers denote BSs and circular markers denote UTs. The distances from each UT to its serving BS were computed and are used as inputs to the pathloss model in Task 2.2.

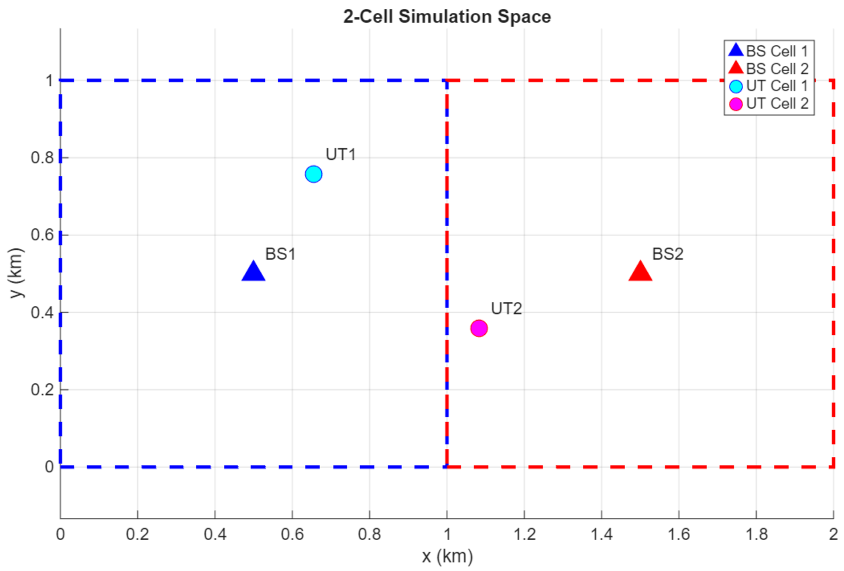

Figure 2: Two-Cell Simulation Space

Task 2.2

Task 2.2(a): Channel Model Implementation

The path loss between each BS-UT pair was computed using the same Urban Micro model from Task 1 as specified in the assignment:

$$PL = 36.7log_{10}(d) + 22.7 + 26log_{10}(fc)$$

In the assignment, $fc = 2.4\ GHz$ is the carrier frequency. Shadow fading is modelled as an independent log-normal random variable per subchannel with standard deviation $\sigma = 4\ dB$. The total channel power gain on subchannel n from BS j to UT k is:

$$H^2(j,k,n) = 10^{-(PL(j,k) + X\_\sigma(j,k,n))/10)}$$

This was computed for all four BS-UT combinations across N = 5 subchannels, producing the gain matrix $H^2(j,k,n)$ used as input to Task 2.2b.

Task 2.2(b): Iterative Multi-User Water Filling Algorithm

Power allocation was performed using the Modified Iterative Water-Filling Algorithm proposed by Yu (2007), which accounts for inter-cell crosstalk interference. The optimisation problem is defined in Eq.(15) by Yu, which maximises the weighted sum rate across all users and subchannels subject to per-BS power constraints. Equal user weights $\alpha k = 1$ were assumed as no weights were specified. Power was initialised equally across subchannels as $P(k,n) = Pmax/N = 2W$ per subchannel. Taking the Lagrangian dual of this problem (Eq.17) and applying KKT conditions yields the modified water filling power allocation (Eq.22):

$$P_k(n) = \left(\frac{\alpha_k}{\lambda_k + t_k(n)} - \frac{I_k(n) + \sigma^2}{|\ H_{kk}(n)\ |^2}\right)^+$$

where λ_k is the Lagrange multiplier (water level) for user k, t_k(n) is the crosstalk penalty term (Eq.19), I_k(n) is the interference received at UT k on subchannel n (Eq.20), and (0,·)⁺ denotes max(0,·).

The crosstalk penalty tk(n) captures the marginal cost of user k's transmission to the other users' rates. It is large on subchannels where user k causes strong interference to others, lowering the effective water level and reducing power allocation there. When tk(n) = 0 the expression reduces to classical single-user water filling, where each user maximises only their own rate with no regard for others.

Algorithm Structure: The algorithm structure is summarised in Yu's report (Yu, 2007, p418).

Both tk(n) and Ik(n) are initialised to zero. On the first iteration, as mentioned, this reduces Eq.(22) to standard single-user water filling, with interference and taxation terms building up progressively through subsequent iterations. The water level λ_k satisfying the power constraint (Eq.23) was obtained via bisection. The full algorithm (Yu, 2007) consists of two nested loops:

1. Inner loop: Fix tk(n). For each user k, compute $I_k(n)$ via Eq.(20), find $\lambda_k$via bisection on Eq.(23), update $P_k(n)$. Repeat until $P_k(n)$ converges.
2. Outer loop: Update $t_k(n)$ via Eq.(19) using the current power allocations. Repeat until $t_k(n)$ converges.

Convergence was declared when the maximum absolute change in any Pk(n) fell below 10^{-8} W, achieved within approximately 20 outer iterations. Both BSs converged to a total allocated power of 10 W, confirming the power constraint is satisfied. As noted by Yu (2007), since Eq.(15) is nonconvex the algorithm is not guaranteed to find the global optimum. It converges to a KKT point which is at least a local optimum. The nonconvexity arises because users are coupled through interference. Improving one user's allocation changes the interference experienced by all others, creating multiple local optima.

Task 2.3

Task 2.3(a): Sum Rate Per User

The achievable sum rate for each user was computed using Shannon's capacity formula (Rappaport, 2002), applied per subchannel:

$$R_k = \sum_{n=1}^{N} \alpha_k \log_2(1 + \mathrm{SINR}_k(n))[\mathrm{bits/s/Hz}]$$

where the SINR at UT k on subchannel n accounts for inter-cell interference:

$$\mathrm{SINR}_k(n) = \frac{P_k(n) \mid H_{kk}(n) \mid^2}{\sigma^2 + I_k(n)}$$

and $I_k(n) = \sum_{j \neq k} P_j(n) \mid H_{jk}(n) \mid^2$ is the interference from the other BS using the final converged power allocation. The results for both users are presented in Table 1.

User	Sum Rate (bits/s/Hz)	Per-subchannel rates (bits/s/Hz)
User 1	19.8832	5.543, 4.933, 0.000, 4.631, 4.776
User 2	32.1203	4.649, 4.450, 10.528, 6.208, 6.286
Network	52.0035	—

Table 1: Sum Rate Per User and Subchannel

User 2 achieves a higher sum rate of 32.12 bits/s/Hz compared to User 1's 19.88 bits/s/Hz, reflecting User 2's shorter distance to its serving BS (252.10 m vs 467.85 m) and consequently stronger direct channel gains. Notably, User 1 allocates zero power to subchannel 3 — the heavy power concentration by User 2 on that subchannel raises the interference floor sufficiently that the $(\cdot)^{+}$ operator in Eq.(22) drives the allocation to zero, with power redistributed to subchannels 2, 4 and 5. Both BSs utilise their full power budget of 10 W, confirming the power constraint is satisfied. The cross-link distances of approximately 1200 m are significantly longer than the direct links, resulting in relatively low inter-cell interference.

Task 2.3(b): Power Allocation Results

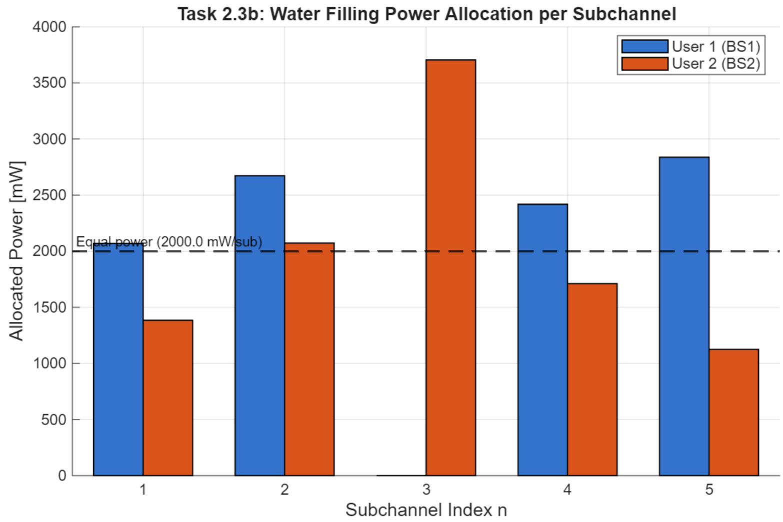

Figure 3

Figure 3 shows the power allocated by each BS across the five subchannels after convergence of the iterative water filling algorithm. Several observations can be made.

Power is redistributed away from equal allocation. The dashed reference line shows what equal power ($P_{max}/N = 2$ W per subchannel) would be. Water filling departs significantly from this, allocating more power to stronger subchannels. User 2 concentrates power on subchannel 3 where its direct channel gain $|H_{22}(3)|^2$ is strongest, achieving 10.528 bits/s/Hz on that subchannel alone. User 1 abandons subchannel 3 entirely in response, redistributing power to subchannels 2, 4 and 5 where interference from User 2 is weaker. The two users allocate asymmetrically reflecting their different distances to serving BSs and different cross-channel conditions.

The modified algorithm of Yu (2007) improves on standard water filling by including the penalty term tk(n), which stops a BS from transmitting heavily on subchannels where it causes strong interference to the other user. Without this term each user would simply maximise its own rate with no regard for the other. The taxation term forces a degree of cooperation.

Yu notes that because the problem is nonconvex, the algorithm only finds a local solution rather than the guaranteed global optimum (Yu, 2007). In practice both users fully use their 10W budget and the network sum rate of 52.00 bits/s/Hz reflects the channel conditions well, suggesting the local solution found is reasonable. This is consistent with Yu (2007, p.419) who notes that in practice the local optimum obtained appears fairly close to the global optimum for similar scenarios.

Task 3: Performance Evaluation of CSMA/CA in IEEE 802.11 Wireless Networks

Task 3.1: CSMA/CA Performance Analysis in Basic Access Mode

Task 3.1.1 — Initialisation of Physical Layer Parameters

The physical layer parameters for the IEEE 802.11a standard were obtained from Clause 17 of the IEEE 802.11a-1999 standard (IEEE, 1999). The following values in Table 2 were used in the Matlab script:

Parameter	Value	Standard Reference	Description
SYMBOL	4 µs	Table 79, Section 17.3.2.3	OFDM symbol interval (TSYM = TGI + TFFT)
PHY_PREAM	16 µs	Table 79, Section 17.3.2.3	PLCP preamble duration
SERVICE	16 bits	Section 17.3.5.1	PHY service field overhead bits
TAIL	6 bits	Section 17.3.5.2	Convolutional encoder reset bits

Table 2: Parameter Values

SYMBOL (T_SYM = 4 µs): The fundamental time unit of transmission, defined as T_GI + T_FFT (IEEE, 1999, Table 79). All data after the preamble is transmitted in discrete symbol intervals, so the total number of symbols determines overall frame duration.

PHY_PREAM (T_PREAMBLE = 16 µs): The first part of every frame, defined as T_SHORT + T_LONG (IEEE, 1999, Table 79). Sent before any data to allow the receiver to synchronise. Included in both Ts and Tc as it precedes every transmission including ACKs.

SERVICE (16 bits): A fixed 16-bit overhead field sent before the data payload (IEEE, 1999, Section 17.3.5.1). Carries no useful data but increases total frame length, affecting the number of OFDM symbols required.

TAIL (6 bits): Six zero bits appended at the end of every frame to reset the convolutional encoder (IEEE, 1999, Section 17.3.5.2). It adds to total frame length and therefore affects symbol count.

These four parameters are essential for computing the actual frame transmission durations used in Bianchi's throughput model, specifically in the calculation of Ts and Tc in equation 14 of Bianchi (2000).

Task 3.1.2 — Transmission Duration for a Successful Access

The successful transmission time T_S is calculated as:

T_S = T_DATA + SIFS + T_ACK + DIFS + PROP;

This implements Bianchi's equation 14 (Bianchi, 2000) for the basic access mechanism, where a successful transmission occupies the channel for the full data frame, followed by a SIFS gap, the ACK response, a DIFS gap, and propagation delay.

T_DATA and T_ACK are computed using the 802.11a OFDM frame structure provided in the template (IEEE, 1999), with data and ACK frames transmitted at 54 Mbps and 24 Mbps respectively, giving different symbol counts.

Task 3.1.3 — Transmission Duration for a Collided Access

The collision time T_C is calculated as:

T_C = T_DATA + DIFS + PROP

This is shorter than T_S because when a collision occurs, no ACK is ever returned. The channel is therefore only occupied for the data frame itself, followed by DIFS and propagation delay, before stations begin their backoff procedures. This directly implements Bianchi's equation 14 for the collision case, and corresponds to the "T collision basic access" shown in Figure 5 of Bianchi (2000).

Task 3.1.4 — Effective Throughput Calculation

Section (4), Collision Probability: The collision probability p cannot be solved analytically as it is implicitly defined through a nonlinear system of two coupled equations from Bianchi (2000). Specifically, Equation (7) defines the transmission probability τ as a function of p, and Equation (9) defines p as a function of τ. Since τ depends on p and p depends on τ, neither can be solved directly. To resolve this, Equation (7) is substituted into Equation (9), reducing the system to a single equation in one unknown p. This equation is rearranged to equal zero:

```
matlabeq = @(p) p - (1 - (1 - (2*(1-2*p)) / ((1-2*p)*(W+1) + p*W*(1-(2*p)^m)))^(n-1));
```

```
p = fsolve(eq, 0.1);
```

The anonymous function matlabeq takes p as its input and computes the difference between the left and right hand sides of Equation (9), with τ substituted from Equation (7). MATLAB's fsolve then finds the value of p that makes this expression equal to zero, starting from an initial guess of p = 0.1, which represents a reasonable collision probability. Once p is found, τ is extracted explicitly using Equation (7), and the remaining quantities P_tr, P_s and throughput S are computed in sequence.

Section (5), Throughput: Throughput is evaluated using Bianchi's Equation (13):

```
matlabBianchi(j) = (Ps * Ptr * EP) / (Ps * Ptr * T_S + Ptr * (1-Ps) * T_C + (1-Ptr) * SLOT);
```

This is computed for each station count j and repeated for all three parameter combinations from Table I in the Briefng, with results plotted on a single graph.

Task 3.2: Impact of m and CW_{min}

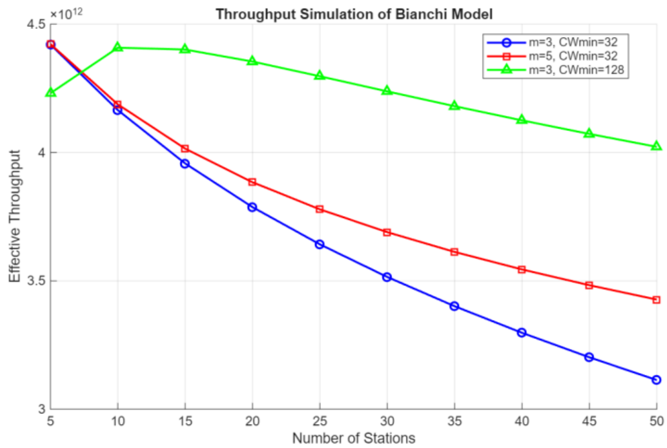

Figure 4: Throughput Simulation

Figure 4 shows the effective throughput as a function of the number of contending stations for three parameter combinations: (m=3, CWmin=32), (m=5, CWmin=32), and (m=3, CWmin=128), all at 54 Mbps.

Observation 1: Throughput decreases with station count for all configurations. As the number of stations increases, the collision probability rises, which wastes channel time. This is consistent with Calì et al. (2000), who showed analytically that throughput degrades with the number of contending nodes under saturation conditions. Their Figure 2 demonstrates this clearly, with the separation between curves for M=10, M=50, and M=100 showing progressively lower capacity (throughput) as station count increases. The underlying mechanism is stated explicitly in their Section II-B: the capacity decreases when M increases due to the increase in collision probability, as the backoff mechanism does not take into consideration the number of active stations. In out Figure 4, the blue curve (m=3, CWmin=32) drops most steeply from approximately 4.38×10^{12} at 5 stations to 3.13×10^{12} at 50 stations. This is a reduction of nearly 29%.

Observation 2: Larger CWmin has the greatest impact on throughput. The green curve (m=3, CWmin=128) mostly outperforms both other configurations, dropping only 4% across the full station range. This relatively flat behaviour demonstrates that a larger initial contention window effectively limits collision probability even as network density increases. Calì et al. (2000) support this directly — their Figure 3 shows that the optimal contention window grows larger as M increases, meaning CWmin=128 is better aligned with the optimal operating point at high load than CWmin=32, which must undergo several collision-triggered doubling stages to reach a comparable window size. The gap between green and blue widens from approximately 0.16×10^{12} at 5 stations to 0.92×10^{12} at 50 stations, confirming that CWmin becomes increasingly important under heavy load.

Observation 3: Increasing m improves throughput under high load but less so than CWmin. Comparing (m=3, CWmin=32) with (m=5, CWmin=32), the red curve (m=5) maintains better throughput as station count grows, dropping from 4.38×10^{12} to approximately 3.43×10^{12} compared to the blue curve's drop to 3.13×10^{12}. A higher m allows the contention window to double more times after repeated collisions. Bianchi (2000) shows this directly reduces collision probability at high load. However, the improvement is smaller than that achieved by increasing CWmin. Calì et al. (2000) explain that increasing m only extends the reactive range of the backoff algorithm, meaning bandwidth is already wasted on collisions before any benefit is realised. Meanwhile, a larger CWmin avoids these initial collisions entirely.

Observation 4: All curves converge at low station counts. At n=5 stations, blue and red begin at approximately 4.38×10^{12} while green starts slightly lower at 4.22×10^{12}. With few competing stations, collisions are rare regardless of backoff parameters, so CWmin and m have minimal effect. The lower starting point of the green curve demonstrates the idle slot penalty of a large initial contention window. Stations wait longer before transmitting even when the channel is uncongested, consistent with Calì et al. (2000) who show in their Figure 3 that large windows waste channel time on idle slots when collision pressure is low. However, by n=10 the green curve overtakes both others as the collision reduction benefit begins to dominate.

Observation 5: No single configuration is optimal across all loads. This motivates adaptive contention window schemes, as discussed in literature Calì et al. (2000), where the contention window is dynamically adjusted based on estimated channel load. The static parameters used here represent a fixed design point that performs differently depending on load.

Task 3.3: Machine Learning Model for Throughput Prediction

Task 3.3.1 — Synthetic Dataset Generation

To generate a training dataset for supervised learning, the Bianchi MATLAB implementation from Tasks 3.1 and 3.2 was used to evaluate throughput across a set of input parameters. For each combination, the effective throughput was computed analytically using Bianchi's model and recorded as the output label. Five parameters were varied to produce the dataset.

n_stations: Values from 5 to 50 in steps of 5 (10 values), capturing network load from lightly loaded to heavily congested. This is the number of contending stations.

m and CWmin: The three combinations from Table I. These are the maximum backoff stage and minimum contention window size respectively.

data_rate_Mbps: All eight IEEE 802.11a PHY rates: 6, 9, 12, 18, 24, 36, 48, and 54 Mbps. ACK rates were assigned as 6 Mbps for 6/9 Mbps data rates, 12 Mbps for 12/18 Mbps, and 24 Mbps for 24/36/48/54 Mbps. This introduces variation in transmission durations T_S and T_C, which directly affect the throughput denominator in Bianchi's Equation (13).

payload_bytes: Three sizes: 1000, 2000, and 4000 bytes, representing small, medium, and large packets. Payload size affects both the data transmission duration T_DATA and the normalisation factor EP = LEN_DATA / SLOT

The **effective throughput** was computed using Bianchi's Equation as in the previous task. This scalar throughput value serves as the output label for supervised learning. The total number of combinations is 720 samples. The dataset was exported to a structured CSV file with the following columns:

Task 3.3.2 — Machine Learning Model Development

A polynomial regression model of degree 2 was chosen to predict throughput from the five input features. Bianchi's throughput formula is inherently nonlinear and throughput does not vary linearly with any of these parameters, so a purely linear model would underfit. A degree-2 polynomial introduces squared terms and cross-products that capture these curved relationships without the interpretability problems of a neural network or the overfitting risk of higher-degree polynomials.

The model was implemented as a three-stage scikit-learn pipeline. **PolynomialFeatures** (degree=2) expands the 5 input features into 21 terms including all pairwise interactions. **StandardScaler** normalises features to zero mean and unit variance. This is necessary because polynomial terms span very different numerical ranges. **LinearRegression** then fits coefficients on the transformed features via ordinary least squares. The 720-sample dataset was split 80/20, giving 576 training and 144 test samples.

	R squared	**MSE**
Training	0.9931	7.48×10^{20}
Test	0.9932	7.88×10^{20}

Table 3: Training and test set simulation results

The test R^2 of 0.9932 indicates the model predicts over 99% of throughput variance accurately on unseen data. The MSE values appear large due to throughput being on the order of 10^{11}, meaning squared errors are naturally very large. The test R^2 being marginally higher than training is not a concern and is consistent with the random 80/20 split.

Feature	Coefficient	Interpretation
data_rate_Mbps	$+9.35 \times 10^{11}$	Higher PHY rate directly increases throughput. Shorter frame transmission times improve channel efficiency.
data_rate × payload	$+4.74 \times 10^{11}$	Higher rates are more beneficial with larger payloads, as more useful data is delivered for the same protocol overhead cost.
payload_bytes²	-4.35×10^{11}	Diminishing returns. Very large payloads offer less additional efficiency gain.
payload_bytes	$+3.51 \times 10^{11}$	Larger payloads spread fixed overhead across more data, improving efficiency.
data_rate²	-3.46×10^{11}	Diminishing returns at very high rates as overhead becomes proportionally larger.
n_stations	-2.12×10^{11}	More stations cause more collisions, reducing throughput
n_stations × CWmin	$+1.72 \times 10^{11}$	Larger contention window is more beneficial in denser networks where collisions dominate.
n_stations × data_rate	-1.24×10^{11}	High data rates are less effective in dense networks due to increased collision overhead.
CWmin × data_rate	$+1.13 \times 10^{11}$	Larger windows pair well with higher rates to reduce collision impact.

n_stations × m	$+1.00 \times 10^{11}$	More backoff stages become more beneficial as network density increases.

Table 4: Shows 10 largest coefficients by magnitude and interprets them.

The dominant features are data_rate and payload_bytes, which is physically consistent. Throughput is fundamentally determined by how efficiently the channel carries data, and both a higher PHY rate and larger payload reduce the proportion of channel time spent on fixed protocol overhead (preamble, SIFS, DIFS, ACK). The negative n_stations coefficient confirms the expected collision penalty in denser networks.

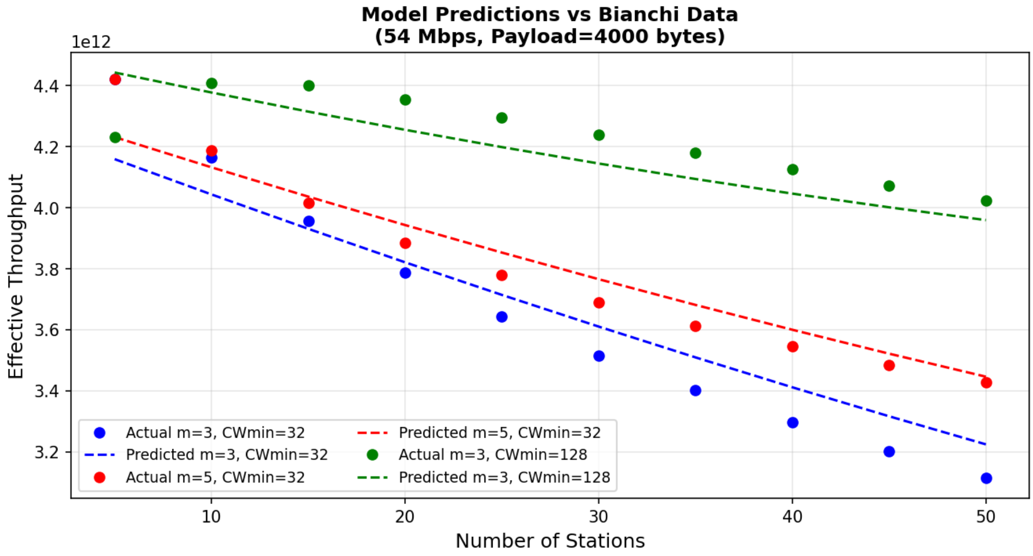

Figure 5

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